

Generation of Computers

01 - Computer Systems and Components

We are living in an Information Society where computers are transforming the way we communicate, do business, and learn. Experts indicate that by the year 2027 every type of job will rely upon getting information, using it, managing it, and relaying information to others.

Computers enable the efficient processing and storage of information. Almost everything we use now contains embedded computers; learning about the use of general-purpose computers, their impact and applications and implications is the focus of this text and this course.

Generations of Computers

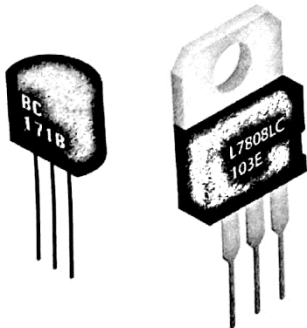
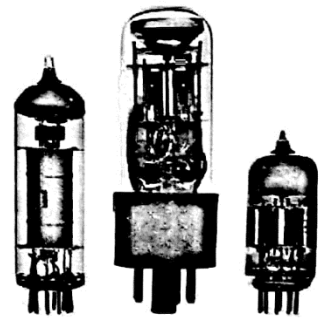
Generation in computer terminology is a change in the technology that a computer was/is using. Initially, the term generation made a distinction between varying hardware technologies; but nowadays, generation includes both hardware and software, which together make up an entire computer system.

There are five computer generations known to date; each generation here is included with some level of detail along with approximate time-period and other characteristics.

First Generation (1942-1952) – Vacuum Tubes

First generation of computer started with using vacuum tubes as the basic components for memory and circuitry for CPU (Central Processing Unit). These tubes like electric bulbs produced a lot of heat and were prone to frequent fusing of the installations, therefore, were very expensive and were affordable only to very large organizations.

In this generation, mainly batch processing operating system was in use. Punched cards, Paper tape, Magnetic tape Input & Output device were in use. There were Machine code and electric wired board languages used.



Second Generation (1952-1964) – Transistors

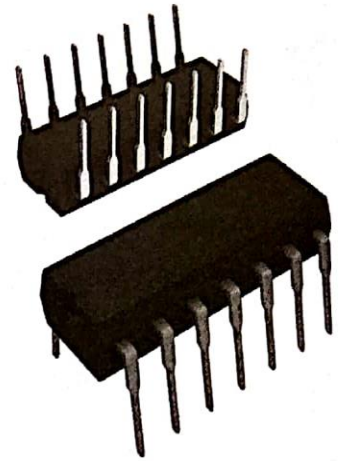
This generation used transistors that were cheaper, consumed less power, more compact in size, more reliable and faster than the first generation machines made of vacuum tubes. In this generation, magnetic cores were in use as primary memory and magnetic tape and magnetic disks as secondary storage devices.

In this generation assembly language and high level programming language like FORTRAN, COBOL was in use. There was Batch processing and Multiprogramming Operating system used.

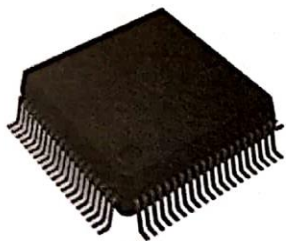
Third Generation (1964-1972) – Integrated Circuits

The third generation of computer saw the use of Integrated Circuits (IC's) in place of transistors. A single I.C. has many transistors, resistors and capacitors along with the associated circuitry; Jack Kilby invented the I.C. This development made computers smaller in size, more reliable and efficient.

In this generation Remote processing, Time-sharing, Real-time, Multi-programming Operating System were used. High-level language (FORTRAN, COBOL, PASCAL, BASIC, ALGOL etc.) were in use during this generation.



Fourth Generation (1972-1990) – Microprocessors



The fourth generation of computers made use of Very Large Scale Integrated (VLSI) circuits. VLSI circuits having about 5000 transistors and other circuit elements and their associated circuits on a single chip made it possible to have microcomputers of fourth generation.

Fourth Generation computers became more powerful, compact, reliable, and affordable. As a result, it gave rise to personal computer (PC) revolution. In this generation Time-sharing, Real time, Networks, Distributed Operating System were used. Other Higher-level languages like C and C++, DBASE etc. were in use during this generation.

Fifth Generation (1990-present) – Artificial Intelligence

In the fifth generation, the VLSI technology became ULSI (Ultra Large Scale Integration) technology, resulting in the production of microprocessor chips having ten million electronic components. This generation focuses on parallel processing hardware and AI (Artificial Intelligence) software.

Fifth generation computing devices, based on artificial intelligence, are still in development, though there are some applications, such as voice recognition, that are being used today. Quantum computation and molecular and nanotechnology will radically change the face of computers in years to come. The goal of fifth-generation computing is to develop devices that respond to natural language input and are capable of learning.

